EMPLOYEE ATTRITION RISK ANALYSIS

Attrition sits at 16.12% and is concentrated in a small set of roles, not across the workforce.

Identifying workforce retention risks across 1,470 employees using Google BigQuery SQL.

1,470
Employees

16.12%
Attrition rate

3
Departments

9
Job roles

OVERTIME

30.53% of employees on overtime leave, versus 10.44% of those who do not.

ROLES

Sales Representatives lose 39.76% of staff, the highest of any role.

EXPERIENCE

Leavers earn 2,046 less per month and stay 2.24 years less.

KEY TAKEAWAY Targeted retention initiatives are likely to deliver greater impact than broad organization-wide programs.

A four-step analysis answers where attrition risk is highest and what drives it.

Business question: which employee groups are most likely to leave, and which factors are associated with that risk?

STEP 1

Dataset

1,470 employee records across 3 departments and 9 roles.

STEP 2

SQL analysis

Attrition rates compared by role, overtime, pay, tenure.

STEP 3

Business insights

Patterns translated into workforce risk findings.

STEP 4

Recommendations

Prioritised actions matched to the largest risks.

BUSINESS IMPLICATION

The analysis shows where attrition risk is highest and which workforce factors contribute most.

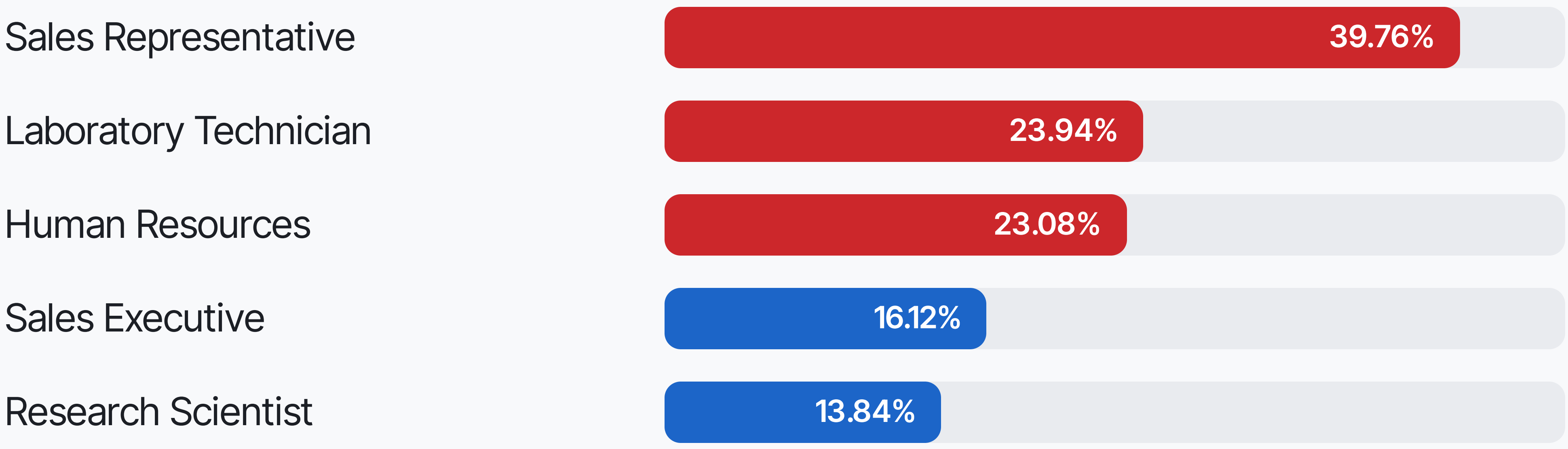
KEY TAKEAWAY

Data-driven analysis enables targeted workforce decisions.

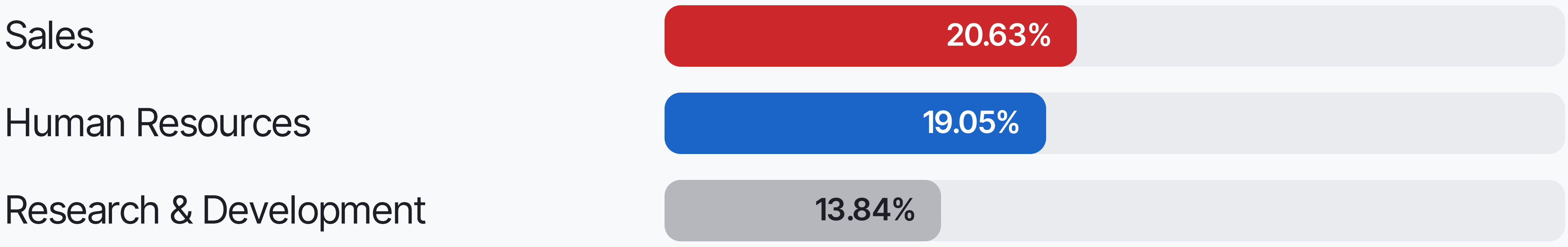
Attrition is concentrated in a few roles, led by Sales.

Role-level attrition ranges from 13.84% to 39.76%, against a 16.12% average.

ATTRITION RATE BY JOB ROLE



ATTRITION RATE BY DEPARTMENT



EXECUTIVE INSIGHT

Three roles carry attrition well above the company average.

Sales Representatives leave at 2.5x the company rate.

Sales is the highest-risk department at 20.63%.

Laboratory Technicians and HR both exceed 23%.

R&D sits below average despite its size.

BUSINESS IMPLICATION

Retention challenges are role-specific rather than organization-wide.

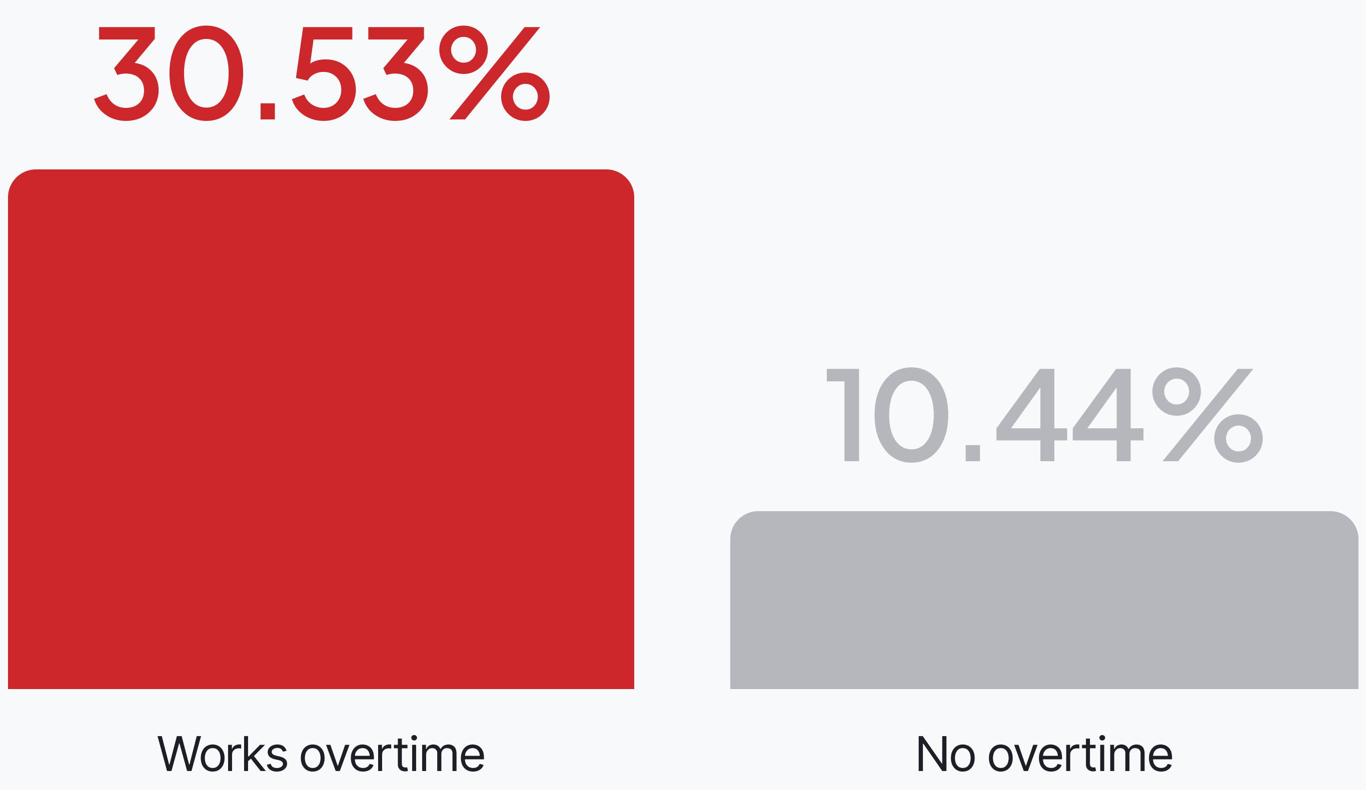
KEY TAKEAWAY High-risk roles should be prioritized for intervention.

Overtime is the strongest single indicator of attrition.

Employees working overtime leave at nearly three times the rate of those who do not.

ATTRITION RATE BY OVERTIME STATUS

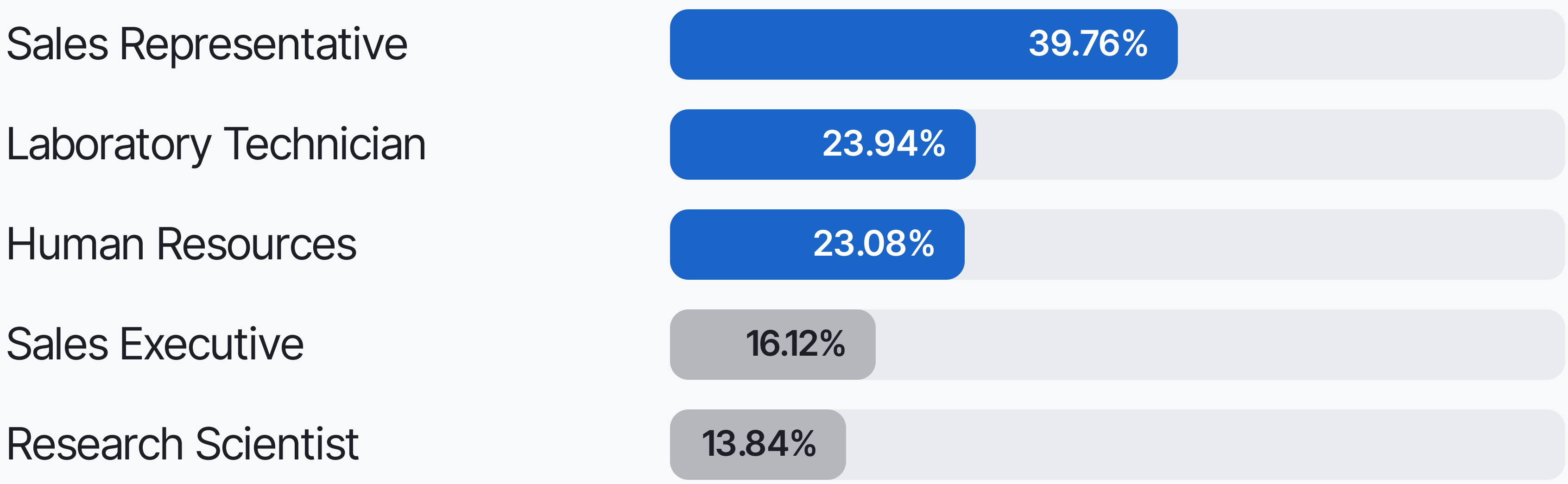
n = 1,470



Gap of **20.09 percentage points**

ATTRITION RATE BY JOB ROLE

n = 1,470



OVERTIME SEGMENT WITHIN TOP ROLE

BUSINESS IMPLICATION

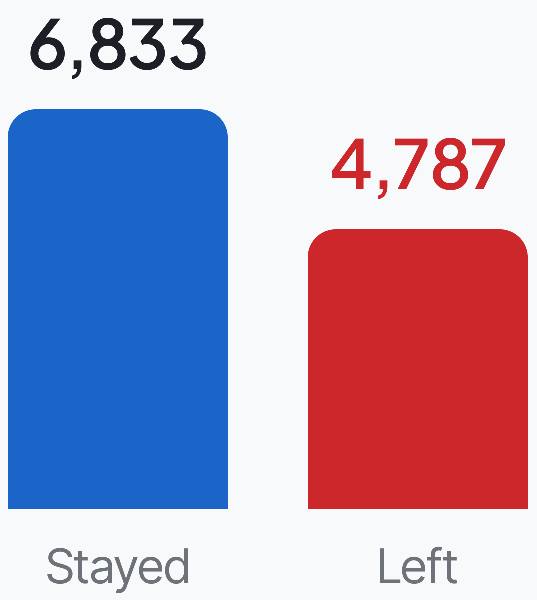
Sustained overtime may significantly increase retention risk.

KEY TAKEAWAY Managing workload is likely one of the highest-impact retention opportunities.

Pay, satisfaction and tenure set stayers apart.

Leavers earn less, report lower satisfaction and have shorter service than those who stay.

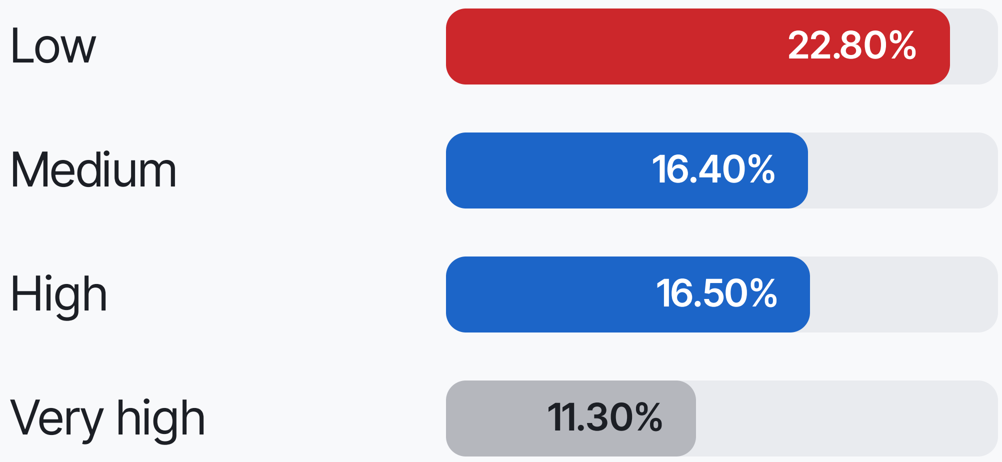
COMPENSATION



Average monthly income is 2,046 lower for leavers.

JOB SATISFACTION

To verify



Directional pattern — confirm rates against the source query.

TENURE

Leavers depart 2.24 years earlier on average.

BUSINESS IMPLICATION

Lower income, lower satisfaction and shorter tenure each align with higher attrition.

KEY TAKEAWAY Retention strategies should address both financial and employee experience factors.

Five connected factors explain most of the attrition risk.

Each factor is supported by evidence from the workforce data reviewed.

Overtime

30.53% attrition with overtime vs 10.44% without.

Workload is the clearest lever available to managers.

High-risk roles

Sales Representatives 39.76%; Sales department 20.63%.

A small group of roles drives a large share of exits.

Compensation

Leavers earn 4,787 vs 6,833 for those who stay.

Pay position affects retention in lower-paid roles.

Job satisfaction

Attrition roughly doubles from highest to lowest satisfaction.

Manager and role quality shape retention outcomes.

Early tenure

Leavers average 5.13 years versus 7.37 years.

Risk is highest in the first years of employment.

COMBINED EFFECT

Employees exposed to several factors at once carry the highest risk of leaving.

KEY TAKEAWAY Attrition is driven by multiple interconnected workforce factors.

Five targeted actions address the drivers identified.

Sequenced by expected impact on attrition and speed of implementation.

01	Reduce excessive overtime	Overtime raises attrition from 10.44% to 30.53%.	Lower exits in the largest at-risk group.	High
02	Prioritize high-risk roles	Sales Representatives leave at 39.76%.	Focused effort on the biggest loss areas.	High
03	Review compensation	Leavers earn 2,046 less per month.	Improved competitiveness in lower-paid roles.	Medium
04	Strengthen onboarding	Leavers average 5.13 years of service.	Better retention through early tenure.	Medium
05	Improve workforce planning	Sales carries 20.63% attrition.	Capacity matched to demand, less strain.	Long-term

KEY TAKEAWAY Targeted interventions are expected to produce greater business impact than broad initiatives.

Workforce data turns retention risk into clear decisions.

WHAT WE LEARNED

- Attrition is 16.12% across 1,470 employees.
- Overtime nearly triples the risk of leaving.
- Three roles carry above-average attrition.
- Pay and tenure separate leavers from stayers.

WHAT LEADERS SHOULD DO NEXT

- Cap sustained overtime in Sales and R&D.
- Launch retention plans for the top three roles.
- Review pay bands for lower-paid roles.
- Extend onboarding through the first two years.

EXPECTED BUSINESS BENEFITS

- Fewer exits in the highest-risk groups.
- Lower hiring and replacement costs.
- More stable sales and laboratory capacity.
- Better employee experience and continuity.

Using workforce data to identify retention risks enables organizations to make more informed, proactive decisions that support employee retention and long-term business performance.

Thank you · Employee Attrition Risk Analysis

Prepared for senior HR and business leadership